

A differentiable, bit-exact game console as a ground-truth testbed for interpretability

The subject IS the substrate — a fully known program — so every explanation can be scored against the true causes.

(1) THE SUBSTRATE

Atari 2600 ROM

joystick do(action)

Differentiable VCS

jutari (JAX) · jaxtari (JAX)

bit-exact to xitari C++ reference

Paper-1: 64 / 64 games (RAM + screen)

✓ Exact forward

deterministic, bit-exact re-run

✓ Exact interventions

do(u) on RAM / register / ROM / input

✓ Full observability

every RAM cell, register, opcode, pixel

✓ Content-path gradients

$\partial y / \partial u$ via SOFT-STE (exact-forward)

(2) THE ORACLE

Exact intervention oracle

— the PRIMARY ground truth —

do(u := v')

occlude · clamp

bit-exact

re-run ROM

$\Delta y(u)$

$= y' - y_0$

true causal map { $\Delta y(u)$ } · no world model assumed

outputs y

+ causes u

cross-check
(1) $\leftrightarrow$ (2) corr.

Content-path gradient companion

$\partial y / \partial u$ via SOFT-STE (raw grad · Integrated Gradients)

CONTENT output

colour / graphics-bit
gradient valid ✓

INDEX output

sprite position (strobe)
naive gradient = 0 ✗

For position / index / event outputs:

the intervention oracle is the SOLE ground truth;
gradient methods are scored AS methods under test.

ground
truth

(3) THE SCORE

any XAI method → explanation $\hat{E}$

saliency · IG · SHAP · lesions · SAE · patching · probes

F Faithful

claimed causes = TRUE causes

precision · recall · sign · magnitude

S Sufficient

predicts y under held-out do(u)

scored vs the exact oracle, not a method

M Minimal

parsimonious, right level

registers / RAM / opcodes; module match

right = F $\wedge$ S $\wedge$ M

verified against the true causes

two reporting axes: faithfulness vs. plausibility

danger zone = looks convincing, isn't faithful

substrate — differentiable VCS (exact forward + interventions)

oracle — exact intervention map $\Delta y(u)$ (primary ground truth)

oracle — content-path gradient companion $\partial y / \partial u$

caveat — index/position output: naive gradient = 0; intervention only

score — correctness triad Faithful $\wedge$ Sufficient $\wedge$ Minimal